CSE413 Speech and Natural Language Processing

Minor Project- Guidelines

1. The objective of the minor project is to enable students to explore current trends in NLP, covering both **research advancements** and **real-world applications**. The students are expected to perform literature review, critical analysis, implementation, and technical communication.

2. **Project Scope and Topic Selection**

   a) Each project shall be based on an application or **recent research paper** in the domain of NLP.

   b) The problems are preferably from the publications in the top venues include (but are not limited to):

   - Conferences: NeurIPS, ICML, ICLR, ACL, EMNLP, NAACL, AAAI, IJCAI, COLING
   - Journals: IEEE TPAMI, JMLR, ACM TIST, Nature Machine Intelligence, IEEE Transactions on Computational Social Systems, Expert Systems with Applications, Information Processing and Management, etc.,

   d) Papers should preferably be published recently.

   e) Students are encouraged to take **problems related to Indian Languages.**

3. **Expected Outcomes**

   a) Thorough understanding and critical analysis of the chosen research paper.

   b) Implementation of the complete method or a **significant component** of the proposed approach.

   c) Comparative analysis with baseline methods or closely related works.

   d) Identification of strengths, limitations, and possible extensions.

4. **Suggested Project Themes**

   See the Appendix-1

5. **Team Formation**

a) Projects may be carried out individually or in groups of up to **four students**.

b) Each team shall submit a **single project proposal, report, and codebase**.

6. **Evaluation Scheme**

The project will be evaluated in **two stages**:

a) **Mid-Term Evaluation (25 Marks)**

i) Problem understanding and literature review.

ii) Methodology/design and clarity of approach.

iii) Progress on implementation.

iv) viva

b) **End-Term Evaluation (25 Marks)**

i) Depth of technical understanding.

ii) Quality of implementation and experimentation.

iii) Comparative analysis and possible extensions.

iv) viva

7. **Contribution to Final Grade**

a) The minor project carries a total weight of **20 marks** toward the final course grade.

b) Marks will be mapped proportionally from the mid-term and end-term evaluation scores.

8. **Important Dates**

a) Team formation and topic submission: **July 31**.

b) Mid-term evaluation: **First week of September**

c) Final report and code submission: **October 16**.

d) End-term evaluation: **October 19-26**.

9. **General Guidelines**

a) Proper citation and plagiarism-free work are mandatory.

b) Use of open-source tools, datasets, and frameworks is encouraged.

c) Code must be reproducible and well-documented.

d) Final submission shall include the notebook along with markdown texts containing problem statements, observations and conclusions. Students can submit the report as a separate PDF, but it is not mandatory.

10. Academic Integrity

Any form of plagiarism, including unacknowledged use of AI-generated text or code, will be treated as per institute academic regulations.

Appendix-1:

List of Topics (but not limited to)

1. Sentiment Analysis in Malayalam/Tamil/Hindi Social Media
2. Fake News Detection for Indian Languages
3. Offensive Language / Hate Speech Detection
4. Code-Mixed Text Classification
5. Language Identification for Code-Mixed Text
6. Automatic Spell Correction for an Indian Language
7. Named Entity Recognition for Indian Languages
8. Part-of-Speech Tagging
9. Morphological Analysis
10. Machine Transliteration
11. News Category Classification
12. Emotion Detection
13. Document Topic Classification
14. NER on Medical Text
15. Character-Level Language Model
16. Next-word prediction using LSTM.
17. Poetry or Lyrics Generation
18. Headline Generation
19. Document Retrieval using TF-IDF
20. Semantic Search using Sentence Embeddings
21. Duplicate Question Detection
22. Extractive Question Answering
23. Reading Comprehension System
24. Restaurant/Library/College Information Assistant
25. Synonym Detection
26. Grammar Error Detection
27. Text Similarity Measurement
28. ASR Error Analysis for Indian Languages

29. Medical Text Classification
30. Legal Document Classification
31. Ayurvedic Text Processing
32. Model Comparison Projects